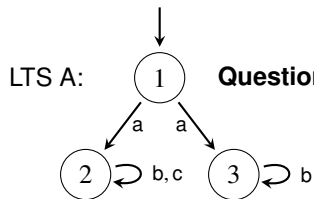
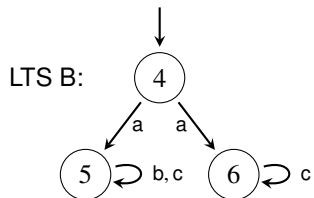


Example: Bisimulation



Question: Is there a bisimulation between A and B, i.e. $A \sim B$?



$\sim_0$	4	5	6
1	1	1	1
2	1	1	1
3	1	1	1

$\sim_1$	4	5	6
1	1	0	0
2	0	1	0
3	0	0	0

$\sim_2$	4	5	6
1	0	0	0
2	0	1	0
3	0	0	0

$\sim_3$	4	5	6
1	0	0	0
2	0	1	0
3	0	0	0

NO! $1 \not\sim 4$

Minimization of Deterministic FA

Given a deterministic and complete FA $A = (S, I, \Sigma, T, F)$

- starting point: $\sim_0 = (F \times F) \cup (\overline{F} \times \overline{F})$

- partition with respect to “final state flag”
- equivalence relation

- refine $\sim_i$ by $\sim_{i+1}$

$s \sim_{i+1} t$ iff $s \sim_i t$ and

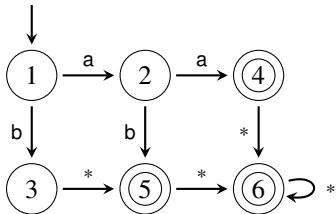
$\forall a \in \Sigma, s' \in S [s \xrightarrow{a} s' \Rightarrow \exists t' \in S [t \xrightarrow{a} t' \wedge s' \sim_i t']]$ and

$\forall a \in \Sigma, t' \in S [t \xrightarrow{a} t' \Rightarrow \exists s' \in S [s \xrightarrow{a} s' \wedge s' \sim_i t']]$

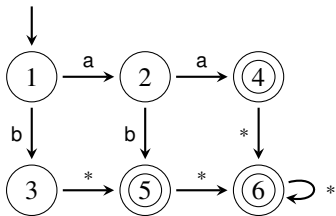
- termination $\sim_{n+1} = \sim_n$ guaranteed at $n = |S|$

- equivalence relation $\sim = \sim_n$ produces minimal automaton $A/\sim$

Example: Minimization

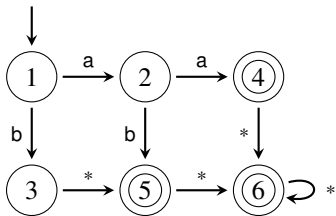


Example: Minimization



$\sim_0$	1	2	3	4	5	6
1	1	1	1	0	0	0
2		1	1	0	0	0
3			1	0	0	0
4				1	1	1
5					1	1
6						1

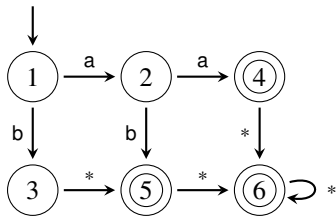
Example: Minimization



$\sim_0$	1	2	3	4	5	6
1	1	1	1	0	0	0
2		1	1	0	0	0
3			1	0	0	0
4				1	1	1
5					1	1
6						1

$\sim_1$	1	2	3	4	5	6
1	1	0	0	0	0	0
2		1	1	0	0	0
3			1	0	0	0
4				1	1	1
5					1	1
6						1

Example: Minimization

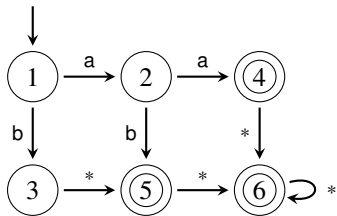


$\sim_0$	1	2	3	4	5	6
1	1	1	1	0	0	0
2		1	1	0	0	0
3			1	0	0	0
4				1	1	1
5					1	1
6						1

$\sim_1$	1	2	3	4	5	6
1	1	0	0	0	0	0
2		1	1	0	0	0
3			1	0	0	0
4				1	1	1
5					1	1
6						1

$\sim_2$	1	2	3	4	5	6
1	1	0	0	0	0	0
2		1	1	0	0	0
3			1	0	0	0
4				1	1	1
5					1	1
6						1

Example: Minimization



$\sim_0$	1	2	3	4	5	6
1	1	1	1	0	0	0
2		1	1	0	0	0
3			1	0	0	0
4				1	1	1
5					1	1
6						1

$\sim_1$	1	2	3	4	5	6
1	1	0	0	0	0	0
2		1	1	0	0	0
3			1	0	0	0
4				1	1	1
5					1	1
6						1

$\sim_2$	1	2	3	4	5	6
1	1	0	0	0	0	0
2		1	1	0	0	0
3			1	0	0	0
4				1	1	1
5					1	1
6						1

